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Frau Dr. Pooja Gupta
Department of Stem Cell Biology
Universitätsklinikum Erlangen
Kussmaulallee 4
91054 Erlangen

Postdoctoral Researcher – AI-based Multi-Omics in Computational Biomedicine

Dear Dr. Gupta,

Three years of experience translating continuous biological models into high-performance Python architectures have prepared me to build the AI frameworks required for your BMFTR-funded project, "AI-Driven Multi-Omics Integration for Predicting IBD-PD Comorbidity Progression". Your group's focus on utilizing machine learning to decode the gut-brain axis and comorbidity networks targets exactly the intersection of complex high-dimensional systems and deep learning that I am passionate about. My transition from computational systems modeling to bioinformatics pipeline engineering makes me a highly suitable candidate to build the AI frameworks your research requires.

While my academic background is in systems modeling, I have spent the last year intensively retraining in modern bioinformatics and machine learning. At my current internship with HealthTwiSt GmbH, I am solely responsible for refactoring a production medical data pipeline (in R/tidyverse) for Germany's interventional radiology quality registry. Working with 143,000 patient records and 1,920 variables, I externalized hardcoded logic into configuration files, verified byte-level reproducibility, and reduced the 1,834-line codebase by 26%, dramatically improving maintainability.

Your project's need for graph neural networks, variational autoencoders, and integrative molecular interaction networks aligns closely with my extensive experience analyzing coupled systems. I bring a strong foundation in the tools you require:

- **AI / Deep Learning:** Built a high-performance Python engine using **Google JAX** with branchless, vectorized array operations for TPU/GPU parallelization—the same paradigms underlying **PyTorch** and **TensorFlow**.
- **Pipeline Development & ML:** Active ML workflows (Tidymodels, Scikit-learn); reproducible bioinformatics pipelines (Nextflow/DSL2) for multi-omic data.
- **HPC & Large-Scale Data:** Processed terabytes of experimental data on HPC clusters; simulation-based perturbation analyses directly applicable to modeling IBD-PD comorbidity networks.

I am deeply motivated by the Universitätsklinikum Erlangen's mission to leverage AI for biomarker discovery and translational impact in complex diseases. My robust mathematical background combined with modern data engineering skills will allow me to immediately contribute to your computational infrastructure. I am available from May 2026 and welcome the opportunity to discuss my candidacy in a personal interview.

Sincerely



Dr. Thejus Mahajan

Enclosures: Curriculum Vitae, Degrees and Certificates